INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	NTT Data Quincy, WA -- station KEPH, America/Los_Angeles
Committed pair	OSM 1343761104 -> 337127824 OSM way 1343761104 / NTT Data Quincy
Facade gap	149.7 m between the two halls
Weather record	42,965 real hourly records from KEPH
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.2993 C at 260 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-04-18 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 1 of 24 hours with 1 mode change. The reactive incumbent took 11 hours -- 10 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 10 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	1 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	11 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 10 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	8.185	18.0	7.780	1.171
01	mechanical	yes	switch budget	8.694	18.0	8.890	1.108
02	mechanical	yes	switch budget	7.555	18.0	6.670	1.071
03	mechanical	yes	switch budget	8.637	18.0	8.890	1.058
04	mechanical	yes	switch budget	7.284	18.0	5.000	1.052
05	mechanical	yes	switch budget	5.883	18.0	4.440	0.936
06	mechanical	yes	switch budget	8.824	18.0	6.254	1.302
07	mechanical	yes	switch budget	9.764	18.0	9.440	1.804
08	mechanical	yes	switch budget	12.540	18.0	13.029	1.927
09	mechanical	yes	switch budget	15.022	18.0	15.560	1.845

10	mechanical	no	dry-bulb	18.391	18.0	18.890	1.834
11	mechanical	no	dry-bulb	20.769	18.0	21.254	1.580
12	mechanical	no	dry-bulb	23.135	18.0	24.440	1.380
13	mechanical	no	dry-bulb	25.609	18.0	27.388	1.215
14	mechanical	no	dry-bulb	26.706	18.0	28.372	1.052
15	mechanical	no	dry-bulb	28.274	18.0	28.890	0.977
16	mechanical	no	dry-bulb	28.673	18.0	28.373	0.948
17	mechanical	no	dry-bulb	28.539	18.0	27.257	1.220
18	mechanical	no	dry-bulb	26.989	18.0	25.003	1.181
19	mechanical	no	dry-bulb	25.336	18.0	22.783	1.354
20	mechanical	no	dry-bulb	23.912	18.0	20.560	1.880
21	mechanical	no	dry-bulb	22.245	18.0	19.440	1.707
22	mechanical	no	dry-bulb	20.307	18.0	17.780	1.377
23	FREE	yes	-	16.689	18.0	12.220	1.255

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.185 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.694 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.555 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.637 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.284 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.883 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.824 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.764 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.540 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.022 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.391 C against a 18.0 C limit, so it fails by 0.391 C. It would take a limit 0.391 C higher, or a bound 0.391 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.769 C against a 18.0 C limit, so it fails by 2.769 C. It would take a limit 2.769 C higher, or a bound 2.769 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.135 C against a 18.0 C limit, so it fails by 5.135 C. It would take a limit 5.135 C higher, or a bound 5.135 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.609 C against a 18.0 C limit, so it fails by 7.609 C. It would take a limit 7.609 C higher, or a bound 7.609 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.706 C against a 18.0 C limit, so it fails by 8.706 C. It would take a limit 8.706 C higher, or a bound 8.706 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.274 C against a 18.0 C limit, so it fails by 10.274 C. It would take a limit 10.274 C higher, or a bound 10.274 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.673 C against a 18.0 C limit, so it fails by 10.673 C. It would take a limit 10.673 C higher, or a bound 10.673 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.539 C against a 18.0 C limit, so it fails by 10.539 C. It would take a limit 10.539 C higher, or a bound 10.539 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.989 C against a 18.0 C limit, so it fails by 8.989 C. It would take a limit 8.989 C higher, or a bound 8.989 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.336 C against a 18.0 C limit, so it fails by 7.336 C. It would take a limit 7.336 C higher, or a bound 7.336 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.912 C against a 18.0 C limit, so it fails by 5.912 C. It would take a limit 5.912 C higher, or a bound 5.912 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.246 C against a 18.0 C limit, so it fails by 4.246 C. It would take a limit 4.246 C higher, or a bound 4.246 C tighter, to change this hour.

22:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.307 C against a 18.0 C limit, so it fails by 2.307 C. It would take a limit 2.307 C higher, or a bound 2.307 C tighter, to change this hour.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.689 C, which is 1.311 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.255 C, and the margin is measured, not chosen: 1.231 C of group-conditional forecast error for this hour of day, plus 0.0241 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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